Abhinay Goud Adhire

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EDUCATION

Master of Science in Computer Science — California State University, San Bernardino *May 2026*

Relevant Coursework: Database Systems, Operating Systems Concepts & Theory, Software Engineering, Theory of Algorithms & Analysis

Bachelor of Technology in Computer Science and Engineering — GITAM University, Hyderabad, India *May 2023*

SUMMARY

Recent Computer Science graduate (M.S., May 2026) with hands-on experience building REST APIs, real-time WebSocket systems, and Kafka/Redis-backed message pipelines. Comfortable across the backend stack — authentication, database design, and distributed message delivery — with a focus on systems that hold up under concurrent use and multi-instance deployment.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, SQL

Backend & APIs: Node.js, Express, RESTful API design, WebSockets (ws)

Messaging & Caching: Apache Kafka (Redpanda), Redis (pub/sub, rate limiting)

Data & Persistence: MySQL, PostgreSQL, Sequelize ORM, relational schema design

Auth & Security: JWT, bcrypt, role-based access control

Tooling: Git/GitHub, Docker & Docker Compose, npm workspaces (monorepo), Postman, Vitest

Foundations: Data structures & algorithms, OOP, operating systems, distributed systems concepts

PROJECTS

Interviewer — Real-Time Technical Interview Platform *Software Engineering Course*

Node.js, Express, TypeScript, MySQL, Sequelize, WebSockets, JWT, bcrypt

GitHub: github.com/abhinay2001/Interviewer

- Built the Express/TypeScript REST API powering a multi-tenant interview platform, including JWT-based auth, bcrypt password hashing, and role-based access for interviewer and candidate accounts.
- Modeled a relational schema in MySQL via Sequelize across 6 linked entities (organizations, users, roles, questions, test cases, sessions) to support scheduling, question banks, and session history.
- Implemented a WebSocket server for real-time collaborative code editing, synchronizing code and language state across concurrent clients with join/leave presence tracking and automatic idle-session cleanup.
- Built a sandboxed code-execution engine (Node's vm module) that runs candidate-submitted JavaScript against configurable test cases under a timeout, returning structured pass/fail results and captured output.
- Added session-level authorization with scheduled time-window enforcement, and an analytics endpoint aggregating completion rate, average session duration, and performance by question difficulty.

Real-Time Chat & Notification Backend *Independent Project*

Node.js, TypeScript, Express, WebSockets (ws), Apache Kafka (Redpanda), Redis, PostgreSQL, Docker

GitHub: github.com/abhinay2001/chat-backend

- Built a multi-room chat backend where messages are durably delivered across multiple backend instances: WebSocket connections publish to Kafka (Redpanda), a consumer writes the durable log to Postgres, and Redis pub/sub fans messages out to whichever instance holds the recipient's connection.
- Implemented a JWT-authenticated WebSocket protocol (join, message, read, unread) backed by a Postgres-enforced room-membership model, with Express handling the REST auth and room endpoints.
- Added Redis-backed presence tracking and rate limiting (5 messages/5 seconds per user), plus a read-receipt system with a forward-only read pointer and per-room unread-count queries.
- Wrote a real end-to-end test suite exercising the live Docker Compose stack (Postgres, Redis, Redpanda) over actual WebSocket connections, alongside in-process API tests — 14 tests total, all passing.